Supplementary Material

The Role of High Mobility Group Box 1 (HMGB1) in Neurodegeneration: A Systematic Review

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SUPPLEMENTARY 1

Table 1. Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Clinical Studies (excluding post-mortem brain samples).

Criteria		(Abu El-Asrar et al., 2013)	(Andersson et al., 2008)	(Asouri et al., 2020)	(Bucova et al., 2020)	(D'Angelo et al., 2018)	(Hamid et al., 2016)	(Malhotra et al., 2015)	(Nasi et al., 2020)	(Sternberg et al., 2016)	(Sternberg et al., 2018)	(Wang et al., 2013)	(Wang et al., 2012)	(Bucova et al., 2019)	(Uzawa et al., 2013a)	(Hwang et al., 2013)	(Festoff et al., 2016)	(Yang et al., 2018)	(Glasnović et al., 2014)
Selection Bias	Q1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Study Design	Type	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Confounders	Q1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Q2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Blinding	Q1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q2	1	1	1	3	1	1	1	1	1	1	3	1	1	1	1	1	1	3
Data Collection Methods	Q1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q2	1	1	1	1	1	1	1	1	1	1	1	1	1	4	1	1	1	1
Withdrawals and drop-outs	Q1	4	4	2	2	2	2	2	2	2	2	2	2	2	1	2	2	2	4
	Q2	1	1	1	1	1	1	1	1	1	1	1	1	1	4	1	1	1	4
Intervention Integrity	Q1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q3	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Analyses	Q1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	Q2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

	Q3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Q4	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Final Rating		**S**	**S**	**M**	**S**	**S**	**S**	**S**	**S**	**S**	**M**	**S**	**S**	**M**	**S**	**S**	**S**	**S**	**S**

Questions

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely 2 Somewhat likely 3 Not likely 4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement 2 60 – 79% agreement 3 less than 60% agreement 4 Not applicable

5 Can't tell

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial 2 Controlled clinical trial 3 Cohort analytic

4 Case-control 5 Cohort (one group pre + post) 6 Interrupted time series

7 Other specify _____ 8 Can't tell

C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes 2 No 3 Can't tell

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most) 2 60 – 79% (some) 3 Less than 60% (few or none) 4 Can't Tell

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes 2 No 3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes 2 No 3 Can't tell

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes 2 No 3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes 2 No 3 Can't tell

F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes 2 No 3 Can't tell 4 Not Applicable

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100% 2 60 - 79% 3 less than 60% 4 Can't tell 5 Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100% 2 60 - 79% 3 less than 60% 4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes 2 No 3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes 5 No 6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation

1 community 2 organization/institution 3 practice/office 4 individual

(Q2) Indicate the unit of analysis (circle one)

1 community 2 organization/institution 3 practice/office 4 individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes 2 No 3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes 2 No 3 Can't tell

SUPPLEMENTARY 2

Table 1. SYstematic Review Centre for Laboratory animal Experimentation Risk of Bias (SYRCLE RoB tool) assessing quality of preclinical animal studies.

Studies	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**Overall**
(Fujita *et al.*, 2016)	U	Y	N	Y	N	Y	N	Y	Y	Y	Unbiased
(Kong *et al.*, 2017)	U	Y	N	Y	N	Y	N	Y	Y	Y	Unbiased
(Meneghini *et al.*, 2013)	U	Y	U	Y	N	Y	N	Y	Y	Y	Unbiased
(Oh *et al.*, 2018)	U	Y	N	Y	N	Y	N	Y	Y	U	Unbiased
(Takata *et al.*, 2004)	U	Y	U	Y	N	Y	N	Y	U	U	Unbiased
(Zhang *et al.*, 2021)	U	Y	U	Y	N	Y	N	U	Y	Y	Unbiased
(Zhang *et al.*, 2020a)	U	Y	N	Y	N	Y	N	Y	Y	Y	Unbiased

(Zhang *et al.*, 2020b)	Y	Y	U	Y	N	Y	N	Y	U	U	Unbiased
(Zhou *et al.*, 2020)	U	Y	U	Y	N	Y	N	Y	Y	U	Unbiased
(Nilson *et al.*, 2017)	U	Y	U	Y	N	Y	N	U	U	U	Unbiased
(Busch *et al.*, 2012)	U	Y	Y	Y	N	Y	N	Y	U	U	Unbiased
(Jang *et al.*, 2013)	U	Y	U	Y	N	Y	N	U	Y	U	Unbiased
(Gan *et al.*, 2020)	Y	Y	U	Y	N	Y	N	Y	U	U	Unbiased
(Kim *et al.*, 2019)	U	Y	U	Y	N	Y	N	U	U	U	Unbiased
(Lv *et al.*, 2019)	U	Y	U	Y	N	Y	N	Y	Y	Y	Unbiased
(Santoro *et al.*, 2016)	Y	U	U	Y	N	Y	N	U	U	U	Unbiased

(Sasaki *et al.*, 2016)	U	Y	Y	Y	N	Y	N	Y	U	U	Unbiased
(Tian *et al.*, 2020)	U	Y	Y	Y	N	U	U	Y	Y	U	Unbiased
(Goula *et al.*, 2009)	U	Y	Y	Y	N	Y	N	Y	U	U	Unbiased
(Lee *et al.*, 2019)	U	Y	U	Y	Y	U	Y	Y	Y	Y	Unbiased
(Lo Coco *et al.*, 2007)	U	Y	U	U	U	U	U	Y	Y	Y	Unbiased
(Meyer *et al.*, 2018)	U	Y	U	U	N	N	Y	Y	Y	Y	Unbiased
(Lee *et al.*, 2020a)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Lee *et al.*, 2015)	U	Y	U	N	N	U	U	Y	Y	Y	Unbiased
(Cunha *et al.*, 2018)	U	Y	U	Y	N	Y	U	Y	Y	Y	Unbiased

(Chu *et al.*, 2021)	U	Y	U	Y	Y	Y	Y	Y	Y	Y	Unbiased
(Djedović *et al.*, 2017)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Latha *et al.*, 2016)	U	Y	U	Y	N	U	U	Y	Y	Y	Unbiased
(Robinson *et al.*, 2013)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Sun *et al.*, 2018)	U	Y	U	Y	N	U	U	Y	Y	Y	Unbiased
(Sun *et al.*, 2015)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Uzawa *et al.*, 2018)	U	Y	U	U	Y	U	Y	Y	Y	Y	Unbiased
(Uzawa *et al.*, 2013b)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Abu El-Asrar *et al.*, 2013)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased

(Busch *et al.*, 2012)	U	Y	U	U	N	N	U	Y	Y	Y	Unbiased
(Böhm *et al.*, 2016)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Ross *et al.*, 2020)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Sakamoto *et al.*, 2017)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Ito *et al.*, 2015)	U	Y	U	Y	Y	Y	Y	Y	Y	Y	Unbiased
(Chang *et al.*, 2014)	U	Y	U	Y	Y	U	Y	Y	Y	Y	Unbiased
(Das *et al.*, 2017)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Das *et al.*, 2019)	U	Y	U	Y	N	Y	U	Y	Y	Y	Unbiased
(Faraco *et al.*, 2007)	U	Y	U	U	N	Y	U	Y	Y	Y	Unbiased
(Qi *et al.*, 2015)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased

(Das *et al.*, 2018)	U	Y	U	Y	N	Y	U	Y	Y	Y	Unbiased
(Gasparotto *et al.*, 2018)	U	Y	U	U	Y	U	Y	Y	Y	Y	Unbiased
(Tajuddin *et al.*, 2018)	U	Y	U	N	N	U	U	Y	Y	Y	Unbiased
(Coleman *et al.*, 2017)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased
(Qin and Crews, 2012)	U	Y	U	U	N	U	U	Y	Y	Y	Unbiased

Abbreviations: N, No; Y, Yes; U, unclear

Questions

1. Was the allocation sequence adequately generated and applied?

2. Were the groups similar at baseline or were they adjusted for confounders in the analysis?

3. Was the allocation adequately concealed?

4. Were the animals randomly housed during the experiment?

5. Were the caregivers and/or investigators blinded from knowledge which intervention each animal received during the experiment?

6. Were animals selected at random for outcome assessment?

7. Was the outcome assessor blinded?

8. Were incomplete outcome data adequately addressed?

9. Are reports of the study free of selective outcome reporting?

10. Was the study apparently free of other problems that could result in high risk of bias?

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported (pg)
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	1-2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	2
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	2
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	2
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	2
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	2
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	2, 4-16 in Tables
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	2, Supplementary Files
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	2
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	2,4-16 in Tables
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported (pg)
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	2
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	2-3
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	2-3
Study characteristics	17	Cite each included study and present its characteristics.	Tables 4-17
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary File
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	NA
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	4-17
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	4-17
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	18-20
	23b	Discuss any limitations of the evidence included in the review.	18-20
	23c	Discuss any limitations of the review processes used.	18-20
	23d	Discuss implications of the results for practice, policy, and future research.	18-21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	21
Competing interests	26	Declare any competing interests of review authors.	21
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71